

Markscheme

Mathematics: analysis and approaches
Practice question bank

167. (a)

Consider the system $x + 2y = 5$ and $2x + 4y = 8$.

Multiplying the first equation by 2 and comparing with the second leads to a contradiction of the form $0 = k$. Find the value of k , and hence state that the system has no solution.

multiply the first equation by 2: $2x + 4y = 10$ (M1)

compare with $2x + 4y = 8$: subtracting gives $0 = 2$ (M1)

$k = 2$, a contradiction, so there is no solution **A1**

[3 marks]